

Data Mining AI&CS&DST402

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Methodology of working within this semester

- 1- the activities of lectures are established as three online quizzes, which are online quizzes.**
- 2- each student is responsible about his mobile, internet for quiz, sending the quiz within internet.**
- 3- each student can take snapshot of quiz because the damage will be happened in internet to save the absence. Because the absence is included in the quiz.**
- 4- if the student doesn't follow the instructions of quiz, his mark will be reduced with the half (writing his name in Arabic and till fourth name, his student code).**
- 5- all quizzes and others will be in the question bank.**
- 6- our credit hours are two and the maximum mark is 100, our marks will be classified as following:**
20 (med term) –40 (practical + activities in lecture) – 40 (final exam)

What is Data Mining?



Mining is the process of extracting useful materials from the earth. It is used to extract minerals, gemstones, oil, gas, and any other materials that are useful or valuable enough to make the process worthwhile. Some examples of substances that are mined include coal, gold, or iron ore. Iron ore is the material from which the metal iron is produced.



DATA: known *Facts* concern about *people, objects, or other entities*. It could be in any format: text, graphics, image ... etc. It is stored on Databases or files.
examples: name, birthdate, job, ... etc

- Collection of records and their attributes
- An attribute is a characteristic of an object
- A collection of attributes describe an object

Objects

Attributes

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Types of Data

□ Record Data

- Transaction Data
- Data Matrix
- Document Data

□ Temporal Data

- Time Series Data
- Sequence Data

□ Spatial & Spatial-Temporal Data

- Spatial Data
- Spatial-Temporal Data

■ Ordered

- Spatial Data
- Temporal Data
- Sequential Data
- Genetic Sequence Data

□ Graph Data

- Transaction Data
- World Wide Web
- Molecular Structures

□ UnStructured Data

- Twitter Status Message
- Review, news article

□ Semi-Structured Data

- Paper Publications Data
- XML format

Record Data

- Transaction Data

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Market-Basket Dataset

Record Data Data Matrix

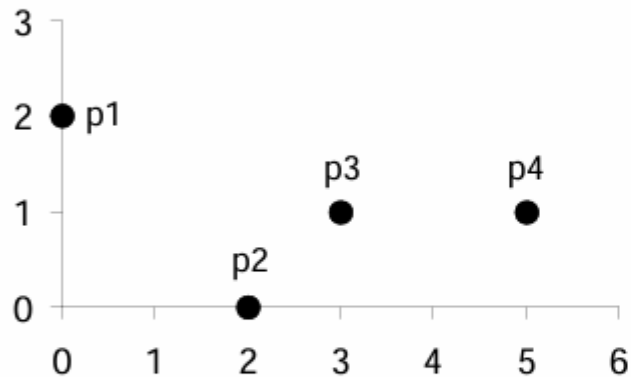
- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- Such data set can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute

Record Data Data Matrix Example for Documents

- Each document becomes a 'term' vector,
 - ▣ each term is a component (attribute) of the vector,
 - ▣ the value of each component is the number of times the corresponding term occurs in the document.

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Record Data Distance Matrix

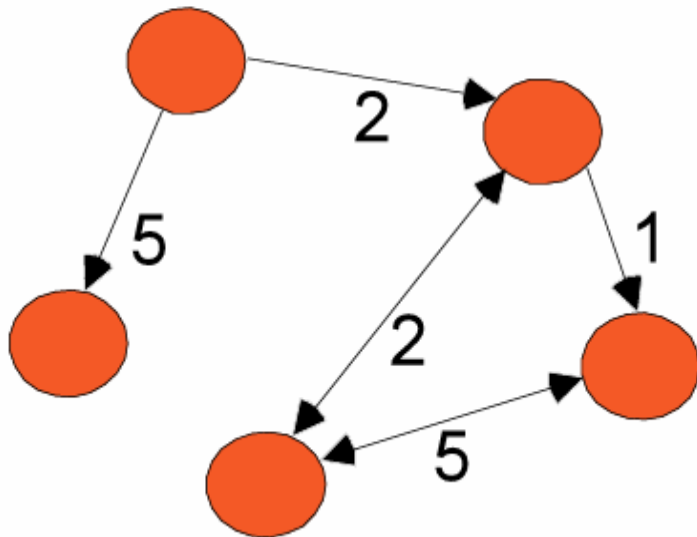


point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Graph Data

■ Examples: Generic graph and HTML Links



```
<a href="papers/papers.html#bbbb">
```

```
Data Mining </a>
```

```
<li>
```

```
<a href="papers/papers.html#aaaa">
```

```
Graph Partitioning </a>
```

```
<li>
```

```
<a href="papers/papers.html#aaaa">
```

```
Parallel Solution of Sparse Linear System of Equations </a>
```

```
<li>
```

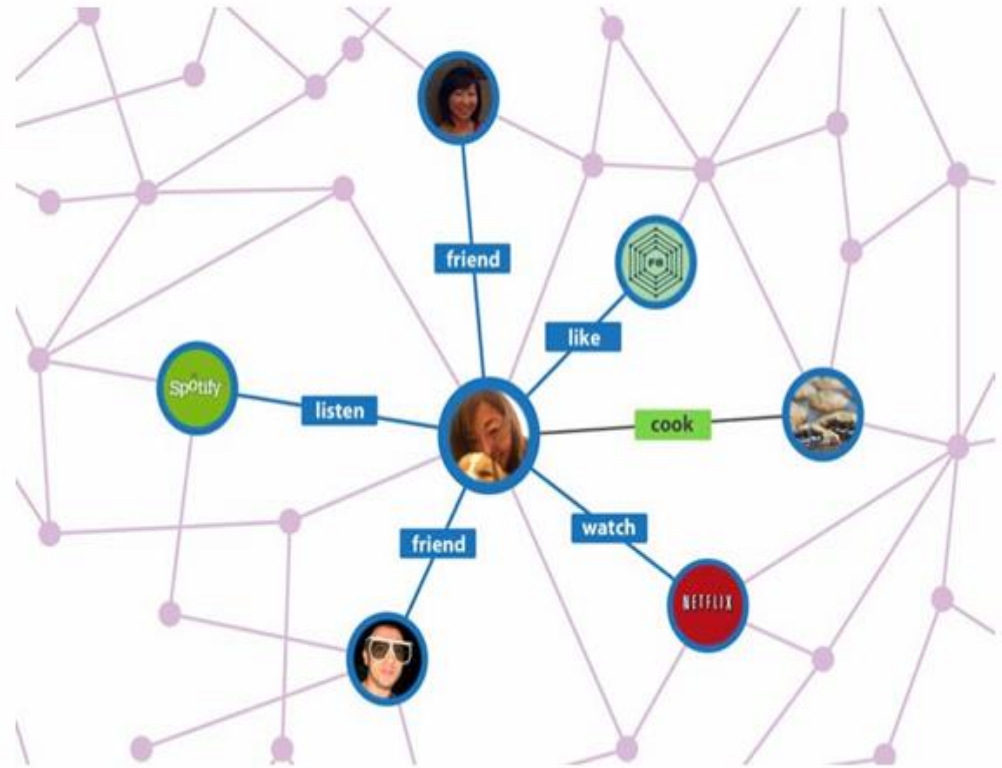
```
<a href="papers/papers.html#ffff">
```

```
N-Body Computation and Dense Linear System Solvers
```

Molecular Structure



Graph Data



Temporal Data

Sequences Data

ID	Symptom Sequence
1	{Night sweat, hypodynamia } →Fever →Achroacytosis→Anemia
2	Night sweat→Fever→Achroacytosis→Anemia
3	Night sweat→Fever→Achroacytosis→Anemia
4	Night sweat→Fever→Achroacytosis→Splénomegalia
5	Night sweat→Fever→Achroacytosis
6	Night sweat→Fever→Anemia
7	Night sweat→Splénomegalia→Anemia
8	Night sweat→Sleepy→Anemia

(Patient Data obtained from Zhang's KDD 06 Paper)

Time Series Data



Yahoo Finance Website

Temporal Data

Biological Sequence Data

Are knesus monkeys or gibbons more closely related to Humans? DNA and polypeptide sequences from closely related species are more similar to each other than are sequences from more distantly related species. In this exercise, you will look at amino acid sequence data for the β polypeptide chain of hemoglobin, often called β -globin. You will then interpret the data to hypothesize whether the monkey or the gibbon is more closely related to humans.

How Such Experiments Are Done Researchers can isolate the polypeptide of interest from an organism and then determine the amino acid sequence. More frequently, the DNA of the relevant gene is sequenced, and the amino acid sequence of the polypeptide is deduced from the DNA sequence of its gene.

Data from the Experiments In the data below, the letters give the sequence of the 146 amino acids in β -globin from humans, rhesus

monkeys, and gibbons. Because a complete sequence would not fit on one line here, the sequences are broken into three segments. The sequences for the three different species are aligned so that you can compare them easily. For example, you can see that for all three species, the first amino acid is V (valine) and the 146th amino acid is H (histidine).

Interpret the Data

1. Scan the monkey and gibbon sequences, letter by letter, circling any amino acids that do not match the human sequence. (a) How many amino acids differ between the monkey and the human sequences? (b) Between the gibbon and human?
2. For each nonhuman species, what percent of its amino acids are identical to the human sequence of β -globin?
3. Based on these data alone, state a hypothesis for which of these two species is more closely related to humans. What is your reasoning?

Species	Alignment of Amino Acid Sequences of β -globin					
Human	1	VHLTPEEKSA	VTALWGKVN	DEVGGEALGR	LLVVYPWTQR	FFESFGDLST
Monkey	1	VHLTPEEKNA	VTTLWGKVN	DEVGGEALGR	LLLVPYPWTQR	FFESFGDLSS
Gibbon	1	VHLTPEEKSA	VTALWGKVN	DEVGGEALGR	LLVVYPWTQR	FFESFGDLST
Human	51	PDAVMGNPKV	KAHGKKVLGA	FSDGLAHLDN	LKGTFAQLSE	LHCDKLHVDP
Monkey	51	PDAVMGNPKV	KAHGKKVLGA	FSDGLNHLDN	LKGTFAQLSE	LHCDKLHVDP

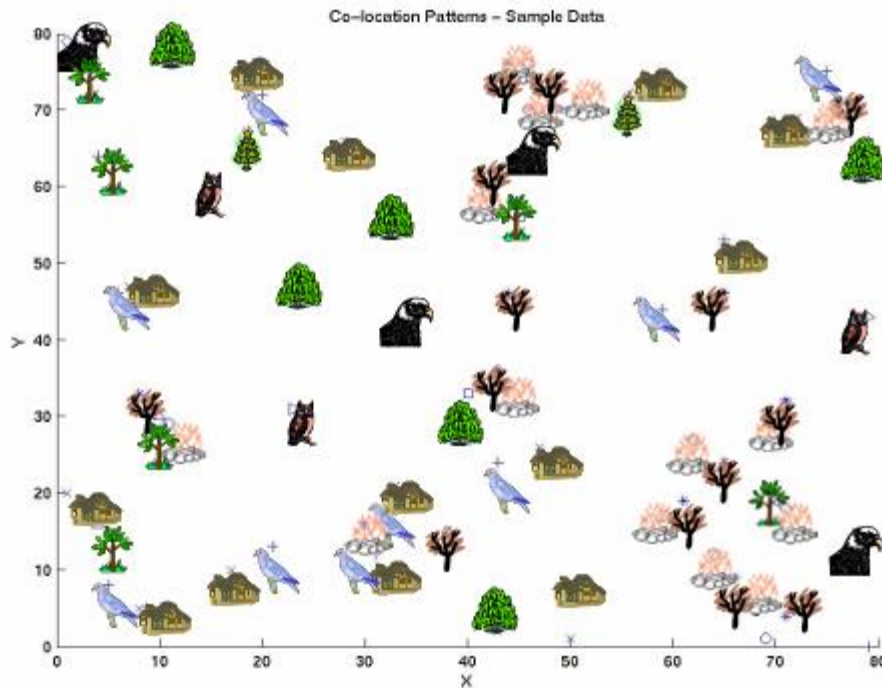
4. What other evidence could you use to support your hypothesis?

 A version of this Scientific Skills Exercise can be assigned in MasteringBiology.

Data from Human: <http://www.ncbi.nlm.nih.gov/ncbi>

Spatial & Spatial-Temporal Data

- Spatial Data

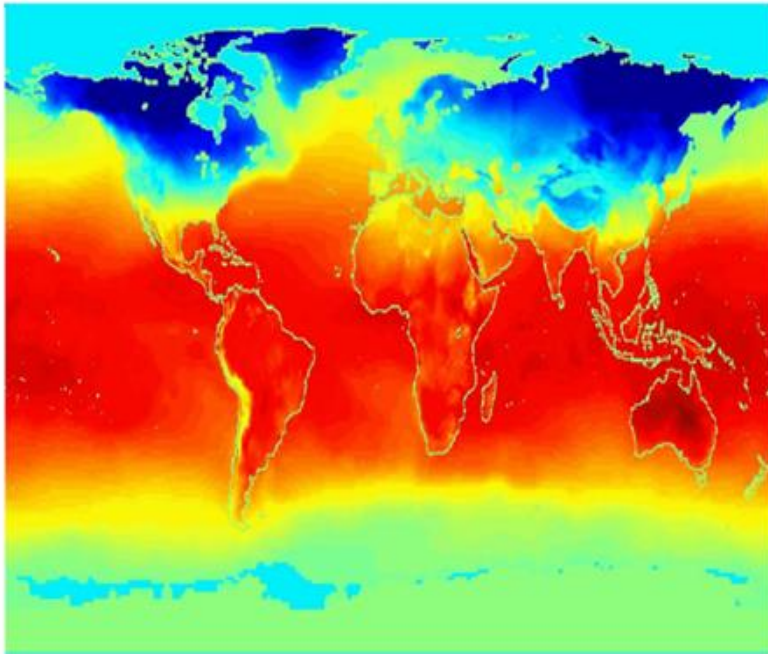


(Spatial Distribution of Objects of Various Types : Prof. Shashi Shekhar)

Spatial & Spatial-Temporal Data

■ Spatial Data

Jan



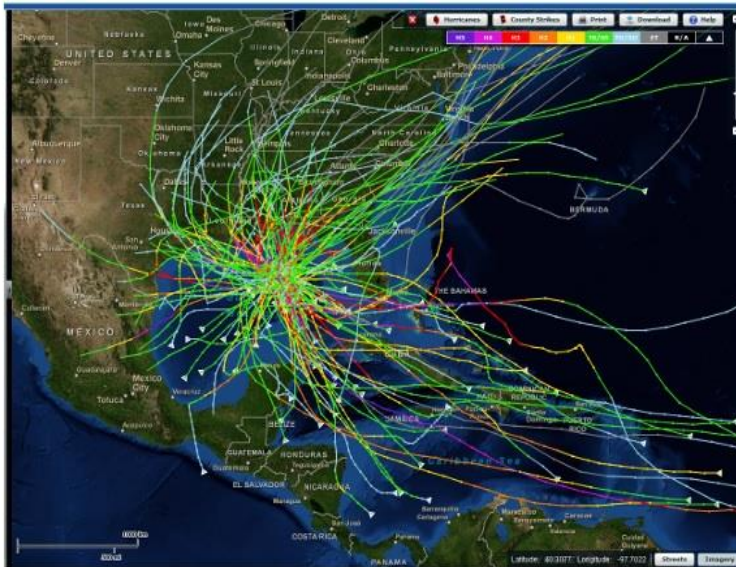
Average Monthly Temperature of land and ocean



Dengue Disease Dataset (Singapore)

Spatial & Spatial-Temporal Data

- Trajectory Data: Set of Harricans



<http://csc.noaa.gov/hurricanes>

- Trajectory Data: (of 87 users obtained using RFID)

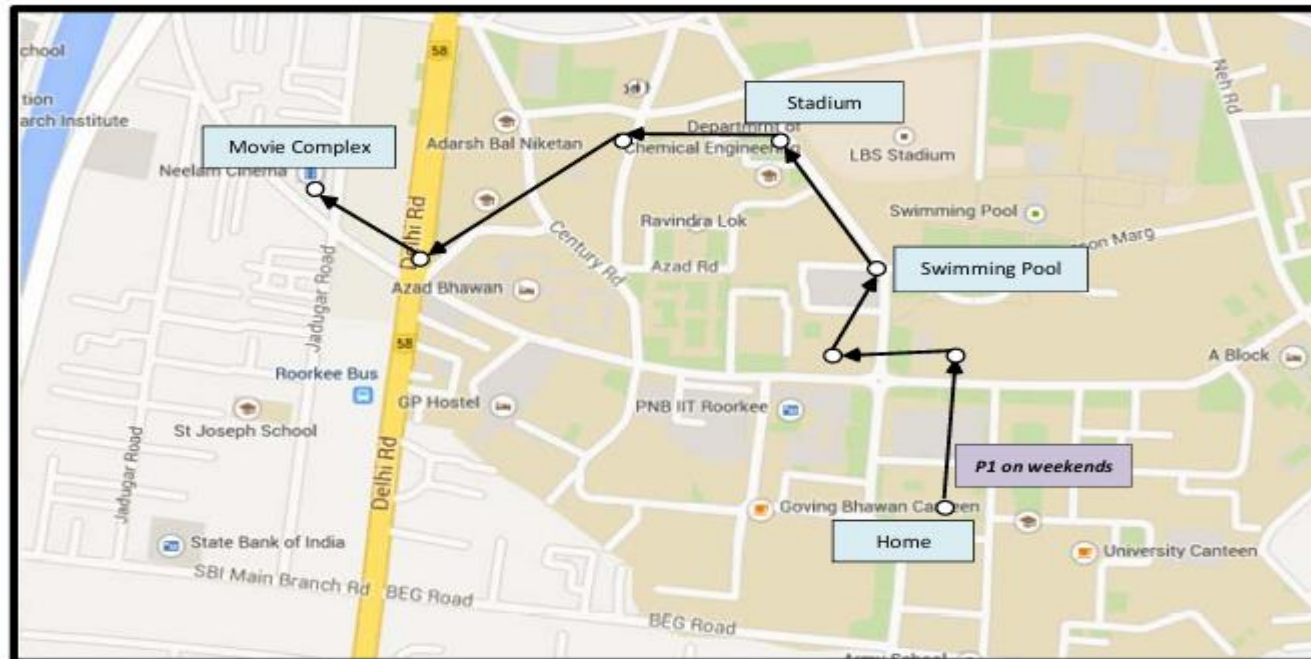


Vast 2008 Challenge – RFID Dataset

Spatial & Spatial-Temporal Data User Movement Data

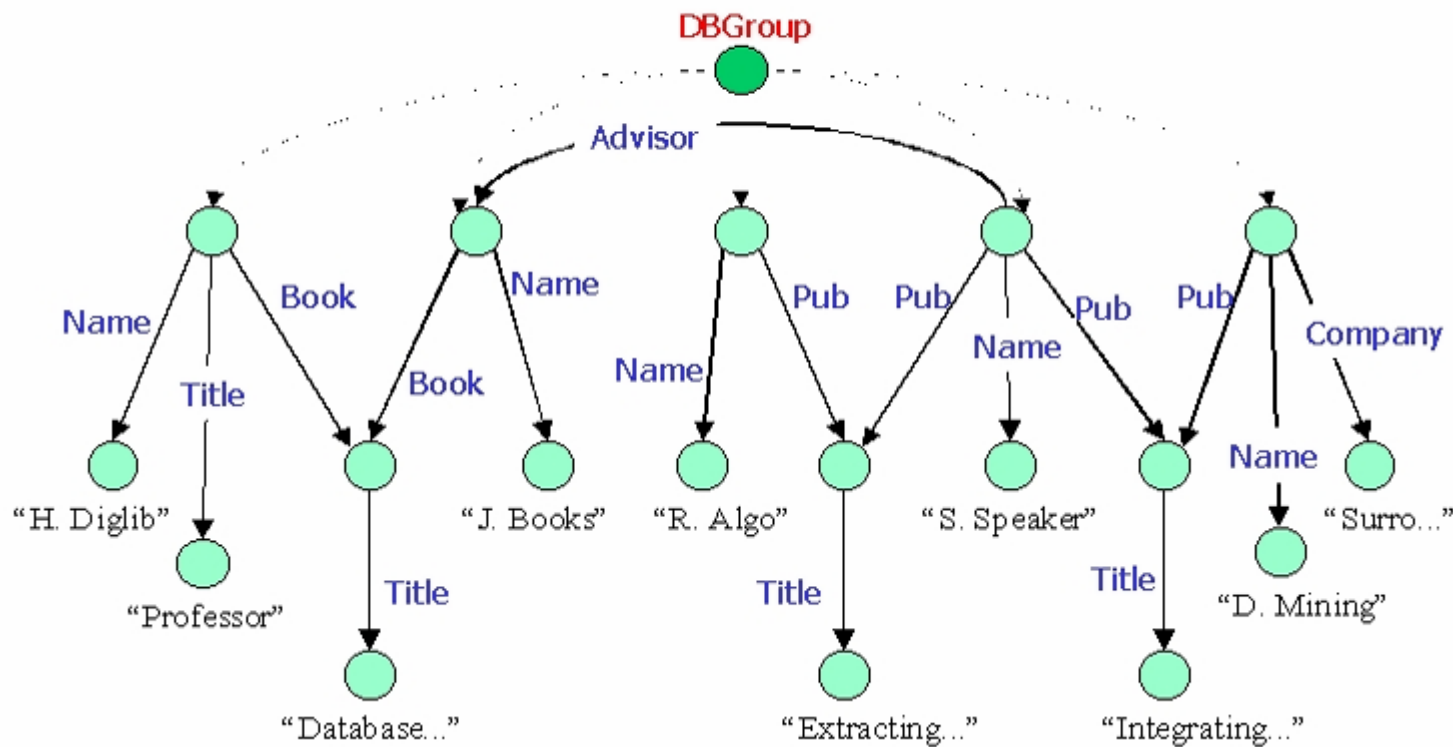
□ Trajectory

- ▣ Movement trail of a user
- ▣ Sampling Points: $\langle \text{latitude, longitude, time} \rangle$



Semi-structured Data

Semistructured Data: Example



Unstructured Data



Jason Y.
Reviews
Written: 69



09/22/2006

Excellent Vietnamese restaurant. Some of my favorites are the sour salmon soup, vietnamese steak, summer rolls, and orange chicken. The tofu w/ginger and scallion special is good. I prefer the pho at pho-specific places. they don't give you basil or chillies with the pho here.

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Aaron E.
Reviews
Written: 75



01/18/2006

They have great fried spring rolls and fresh garden rolls. However, for \$3.50 2 smallish rolls doesn't seem that great. They make up for this with a large bowl of Pho soup for around \$6.00. This place is great for food just before you see a movie at the Uptown theater. However, leave plenty of time. When the place is busy, the waitstaff or kitchen starts to suck and get real slow. You will miss your movie. Give yourself 1.5 hours before a movie start time. This PHO place is not as ghetto as other PHO places (like PHO 75).

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Sean G.
Reviews
Written: 126



03/06/2006

The pho is good -- esp for DC -- but not great compared to some of the Pho joints in Seattle who can drop a bowl of broth and noodles that will blow your mind for under \$4 bucks. Travelling... the downside is you learn what youre missing..... That said, this is a decent spot for a late lunch over a hot bowl of the the good stuff.

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Nils J.
Reviews
Written: 76



04/14/2006

Good pho, reasonable prices, and great for eating dinner before you go to the Uptown for a flick. The wait staff can be very unresponsive, though, which is annoying. Unfortunately, they don't give you basil and cilantro along with the sprouts to dump into your pho.

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Nathalie L.
Reviews
Written: 208



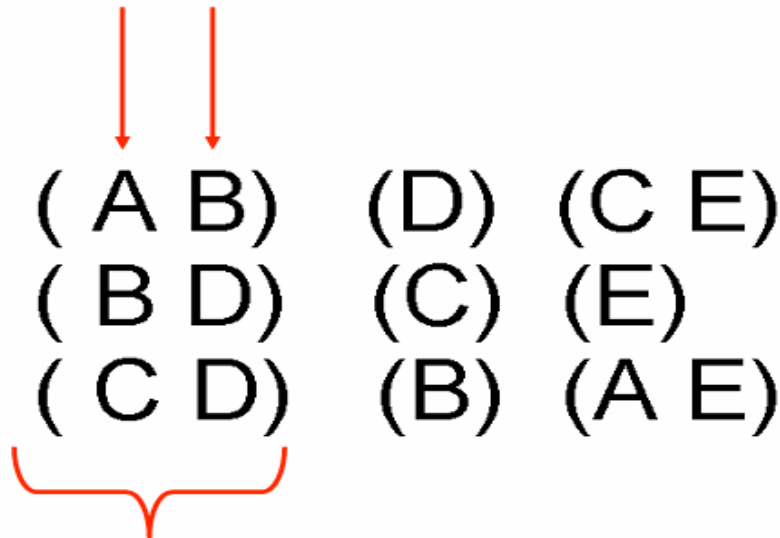
07/08/2006

I was missing home one day and decided to pop in for some comfort food and Nam Viet was exactly what I needed. The waitresses and waiters speaking Vietnamese and the aroma of the pho really lifted my spirits that day.

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Ordered Data

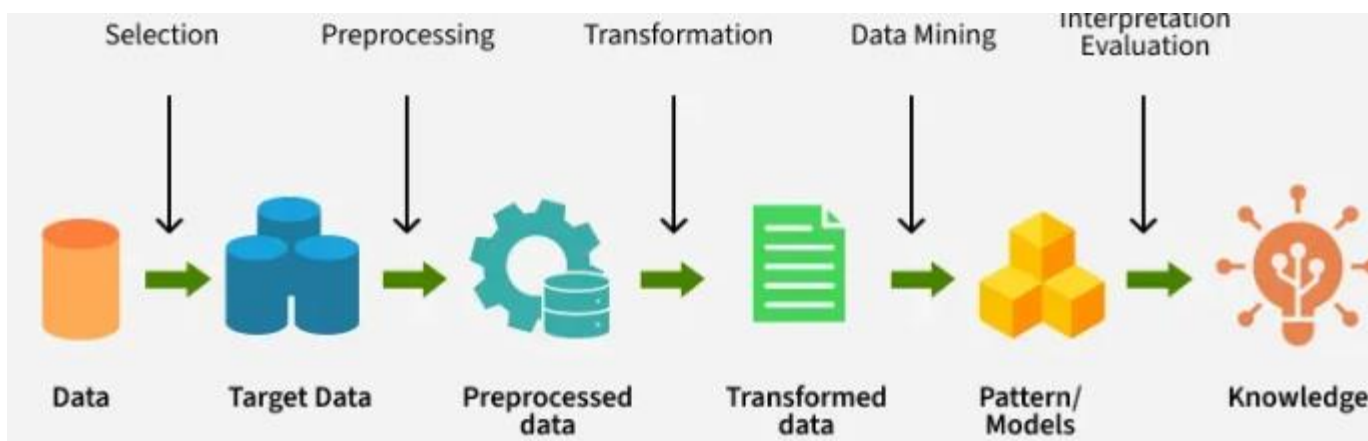
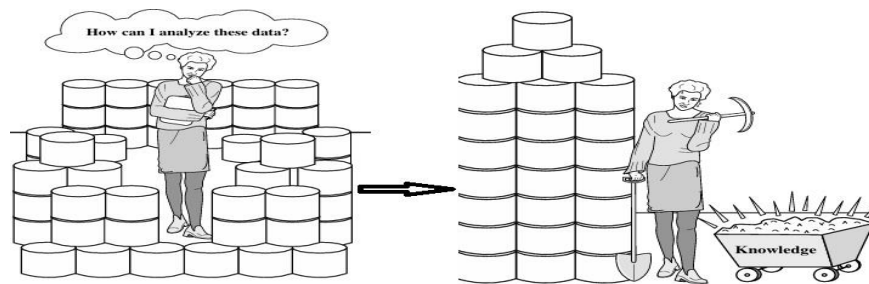
- Sequences of transactions
Items/Events



- Genomic sequence data

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```


Data mining refers to extracting or “mining” knowledge from large amounts of data stored in databases, data warehouses, or other information repositories (flat files, spread-sheet, or relational tables) and data may reside in a centralized data repository or be distributed across multiple sites.

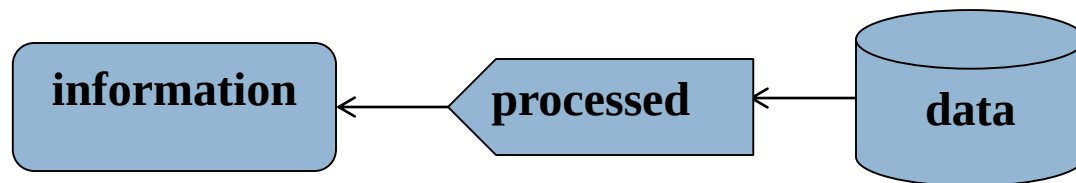


INFORMATION: *it is processed data which is presented in a form suitable for increasing knowledge of the user.*

examples: the students get excellent grade, the employees get salary ≥ 3000 .

Data is converted into information by programs and queries. Data can be defined as “raw material” , information as “intermediate product” and the final product is knowledge.

KNOWLEDGE: *Insights into appropriate actions based on interpreted data.*



Importance of data mining

- Data mining is important because it allows us to discover meaningful patterns and relationships in large volumes of data in a relatively quick and efficient way.
- Volume of information is increasing everyday that we can handle from business transactions, scientific data, sensor data, Pictures, videos, etc. So, we need a system that will be capable of extracting essence of information available and that can automatically generate report, views or summary of data for better decision-making.
- Data mining techniques can take advantage of data coming from different sources like social media platforms or customer databases and convert it into useful insights. In turn, these can answer business or research questions, make predictions, and inform decision making.

Why Data Mining? Commercial Viewpoint

Data mining is used in business to make better managerial decisions by:

- Automatic summarization of data
- Extracting essence of information stored.
- Discovering patterns in raw data.

- Lots of data is being collected and warehoused
 - Web data, e-commerce
 - purchases at department/ grocery stores
 - Bank/Credit Card transactions
- Computers have become cheaper and more powerful
- Competitive Pressure is Strong



Why Mine Data? Scientific Viewpoint

- Data collected and stored at enormous speeds (GB/hour)
 - remote sensors on a satellite
 - telescopes scanning the skies
 - scientific simulations generating terabytes of data
- Traditional techniques infeasible for raw data
- Data mining may help scientists
 - in classifying and segmenting data
 - in Hypothesis Formation



What is (not) Data Mining?

□ What is not Data Mining?

- Look up phone number in phone directory
- Query a Web search engine for information about “Amazon”

□ What is Data Mining?

- Certain names are more prevalent in certain US locations (O’Brien, O’Rourke, O’Reilly... in Boston area)
- Group together similar documents returned by search engine according to their context (e.g. Amazon rainforest, Amazon.com,)

Origins of Data Mining

□ Draws ideas from

- 1- search algorithms, modeling techniques, and learning theories from artificial intelligence, machine learning and pattern recognition
- 2- sampling, estimation, and hypothesis testing from statistics
- 3- database systems, parallel Computing and distributed computing

